Postdoctoral Position (15-18 months)

Representation Learning and Clustering of Interacting Timed Systems

Location : Lacodam Team, IRISA/ INRIA (http://www.irisa.fr), Rennes, France

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Duration: 15-18 months, depending on the starting date
Starting date: As soon as possible from Spring 2026
Keywords: *Neuro-Symbolic AI, Temporal Graph Representation Learning, Timed Automata, Graph Variational Autoencoder, Clustering in Latent Space, Prototype Reconstruction, Behavioral Modeling.*

A- Scientific Context

Recent advances in computer vision and sensor technologies allow the capture of rich, time-resolved information about individual behavior. From video recordings and multi-modal sensors, it is possible to detect behavioral state changes and estimate interactions between individuals in a group. These observations provide the essential data to model both individual and collective dynamics. To achieve interpretable modeling, Timed Automata (TA) can be used as a symbolic representation framework for behavioral states and temporal transitions.
While learning a TA for an individual over a single time period is feasible, applying this approach to long-duration or repetitive behaviors across multiple individuals quickly becomes uninformative and difficult to interpret. The objective of this post-doctoral position is therefore to define a “typical behavior” for each individual, while preserving the temporal and interaction structure between individuals within a group.

B- Objectives

Previous work developed within the team has already proposed a tool for learning timed automata from timed event sequence. The goal of this post-doctoral position is to develop a framework to analyze the behavior of multiple individuals using Timed Automata (TA). This framework will be scalable, interpretable, and generative, and will rely on embedding-based representations. The objective is to build compact representations of these behaviors while keeping important information about interactions and temporal constraints.

Our aim is to build compact and expressive behavioral representations of TA while preserving interactions and temporal dynamics while preserving their interactions and temporal dynamics. A previous approach already enables the learning of TAs from behavioral data over

long periods, which can be reused and further refined to effectively capture interactions between individuals within a group.

Specifically, the objectives are:

1- **Learning compact, latent representations of Timed Automata**
   - Develop methods to embed individual TAs into a continuous latent space.
   - Explore approaches such as Graph Neural Network encoders, variational autoencoders (VAE), or alternative structured representation learning techniques.
   - Ensure that the latent representations capture both state transitions and temporal constraints preserving the behavioral semantics of each automaton.
   - Extend the framework to multi-automata systems while capturing the synchronizations between the other timed automata in interactions.
2- **Clustering and prototype extraction**
   - Perform clustering in the latent space to identify typical behavioral patterns among individuals and groups.
   - Construct prototype automata, providing interpretable summaries of typical individual and collective behaviors.
   - Apply clustering to detect anomalous or rare behaviors, enabling potential applications in monitoring or welfare assessment.
3- **Evaluation and validation**
   - Assess embeddings and clustering in terms of semantic preservation, behavioral interpretability, and interaction fidelity.
   - Compare alternative to identify the most effective approach for structured, temporal, interacting systems.
   - Evaluate on real datasets, including PEPR WAIT4 datasets on interactions between farm animals to estimate their welfare.

C- <u>Expected Impact.</u>

This project is expected to make significant contributions at both the methodological and applied levels.

- Research innovation: development of a novel approach for learning latent representations of symbolic, temporally structured systems such as Timed Automata, which has not been done before.
- Interpretability: automata provide interpretable summaries of individual and collective behaviors.
- Application to animal behavior: facilitates behavioral learning and monitoring of interactions in farm animals (e.g., using WAT4 datasets).
- Broader applicability: framework can be applied to any interacting temporal system, beyond the motivating case study.
- Bridging fields: combines neuro-symbolic AI, graph representation learning, and generative modeling in a unified approach.

D- Candidate's profile

We are looking for a highly motivated candidate with a PhD in Computer Science, preferably with a focus on Artificial Intelligence or Machine Learning.

- The candidate should have strong programming skills and be comfortable working with latent representations, generative models, and structured formalisms.
- The candidate should be able to collaborate with agronomy researchers on applications such as animal welfare
- The ideal candidate is curious, independent, and motivated to contribute to research at the intersection of interpretable AI, representation learning, and interacting temporal systems. He has a strong motivation to contribution to an unexplored but promising domain.

References

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